IDENTIFYING DAMAGED BUILDINGS IN THE PALLISADE WILDFIRE

Using Machine Learning Methods Integrated with GIS

-Satya Venkata Siddhartha Bokka

ABSTRACT

Wildfires have become an increasingly severe threat to ecosystems and human infrastructure, particularly in California, where climate variability and expanding development in wildland-urban interfaces have intensified fire events, such as the Camp Fire (2018), Dixie Fire (2021), and Palisade Fire (2025). The 2025 Palisade Wildfire caused significant structural damage, highlighting the urgent need for rapid, accurate, and scalable damage assessment techniques. This study presents a machine learning (ML) and Geographic Information System (GIS)-integrated framework for building damage classification using high-resolution satellite imagery. Multitemporal 3-meter, 8-band imagery from Planet Explorer was used to derive key spectral indices—Normalized Difference Vegetation Index (NDVI) and Normalized Burn Ratio (NBR)—for pre- and post-fire conditions. From these, delta indices (ΔNDVI and ΔNBR) were calculated to measure vegetation loss and burn severity. Additional geospatial variables such as slope (derived from Digital Elevation Model [DEM]), land cover classification (National Land Cover Database [NLCD]), and wind speed (from the Global Wind Atlas) were included to enhance model performance. Ground truth data for 9,543 buildings were labeled as Damaged or Undamaged and integrated with spatial attributes using ArcGIS Pro.

Four supervised ML models—Random Forest, Support Vector Machine (SVM), Logistic Regression, and Extreme Gradient Boosting (XGBoost)—were trained using an 80/20 stratified split. Among them, XGBoost achieved the highest accuracy (83.2%) with F1-scores of 0.90 (Damaged) and 0.53 (Undamaged). Feature importance analysis revealed ΔNDVI, land cover, slope, wind, and ΔNBR as key predictors of building damage severity in wildfire-affected regions. Final results were visualized using an interactive Folium map with damage predictions overlaid on building footprints. This study demonstrates the value of integrating ML and GIS for spatially detailed wildfire damage assessment and supports the development of operational tools for disaster response and recovery planning.

1. INTRODUCTION

Wildfires are among the most severe and fast-moving natural hazards affecting both ecological systems and built environments. With increasing global temperatures, prolonged drought conditions, and shifts in vegetation patterns due to climate change, the frequency and severity of wildfires have risen dramatically over the past few decades, as emphasized by Singha et al. [3], Moghim and Mehrabi [7], and Thị et al. [8]. Nowhere is this trend more evident than in the western United States—particularly California—which has become synonymous with wildfire-related disasters, as shown in studies by Prince [4] and Ban et al. [12]. Among the most impactful recent incidents, the Palisade Wildfire of 2025 serves as a poignant reminder of the need for rapid, accurate, and scalable post-disaster assessment tools, as supported by Sathishkumar et al. [6].

When wildfires sweep across urban and semi-urban areas, they can destroy thousands of structures, displace residents, and overwhelm emergency response systems. In such scenarios, accurate and timely assessment of building damage is crucial for guiding rescue operations, prioritizing resource distribution, and planning for long-term reconstruction, as described by Galanis et al. [13] and Xu et al. [25]. However, conventional damage assessment methods—such as ground surveys and visual inspections—are often too slow, labor-intensive, and costly to be practical for large-scale disasters, as noted by the Global Facility for Disaster Reduction and Recovery [26]. These methods may take weeks or even months to complete, during which time victims may go without aid, and decision-makers may lack the necessary information to allocate resources efficiently.

The emergence of remote sensing and machine learning (ML) technologies offers a promising alternative. By leveraging satellite imagery, environmental variables, and advanced classification algorithms, it is now possible to automate the process of identifying damaged buildings with considerable accuracy, as demonstrated by Khosravi and Malek [1], Zhang et al. [5], Moghim and Mehrabi [7], Marjani et al. [11], and Zhang et al. [29]. Moreover, when integrated with Geographic Information Systems (GIS), these technologies provide an enriched spatial understanding of the affected areas, enabling decision-makers to visualize patterns, hot spots, and damage severity at both macro and micro scales, as outlined by Ahajjam et al. [10], Li et al. [30], and Kashtan and Hnatushenko [34].

This research project aims to develop and validate a machine learning-based framework integrated with GIS to assess and map building damage caused by the Palisade Wildfire 2025. High-resolution satellite data acquired from Planet Explorer (with 3-meter spatial resolution and 8-band spectral coverage) were used for pre- and post-fire analysis. Spectral indices such as the Normalized Difference Vegetation Index (NDVI) and Normalized Burn Ratio (NBR) were computed to quantify vegetation loss and burn severity, as applied by Prince [4] and Zhang et al. [5]. These indices, along with derived delta metrics (ΔNDVI and ΔNBR), form the core indicators of wildfire impact on land cover.

To supplement spectral features, topographic data was extracted from the USGS GTOPO30 DEM, which was used to compute slope—an important factor influencing fire spread, supported by Singha et al. [3] and Ahajjam et al. [10]. Land cover classification data were sourced from the National Land Cover Database (NLCD) to provide context regarding urbanization and vegetation types. Furthermore, wind speed—a key driver of wildfire propagation—was integrated from the Global Wind Atlas to account for environmental conditions at the time of the incident, as discussed by Moghim and Mehrabi [7] and Ahajjam et al. [10].

One of the most important components of this study is the ground truth data comprising detailed attributes for 9,543 buildings within the affected zone. Each data point includes the building's geographic coordinates, structural type, and damage classification (originally in 5 severity levels, later simplified to binary classes: Damaged and Undamaged). This dataset forms the foundation for training and evaluating ML models and was meticulously preprocessed using ArcGIS Pro for spatial alignment and attribute merging, similar to the approach of Galanis et al. [13] and Gupta et al. [14].

In terms of classification algorithms, four supervised learning models were implemented and evaluated: Random Forest (RF), Support Vector Machine (SVM), Logistic Regression (LR), and Extreme Gradient Boosting (XGBoost). These models were selected based on their robustness, interpretability, and performance in high-dimensional classification tasks, as seen in Zhang et al. [5], Sathishkumar et al. [6], Thị et al. [8], and the XGBoost documentation [42]. Stratified data splitting (80% training, 20% testing) was used to ensure class balance and prevent model bias toward the majority class, following best practices in GeeksforGeeks [41]. StandardScaler was used for feature normalization, and LabelEncoder was applied to transform categorical labels into numerical form for training.

Preliminary results demonstrated that XGBoost outperformed all other models in terms of accuracy, precision, recall, and F1-score. XGBoost's ability to capture non-linear relationships, handle imbalanced data, and rank feature importance makes it particularly suitable for disaster assessment applications, as validated in Zhang et al. [5] and the XGBoost documentation [42]. In this study, XGBoost achieved an accuracy of 83.2%, with balanced F1-scores of 0.90 for Damaged and 0.53 for Undamaged classes, indicating its strong performance in both majority and minority class predictions.

Beyond model accuracy, the practical implementation of this framework is emphasized through interactive visualization using the Folium library. A dynamic map was developed, overlaying building footprints with predicted damage labels. Each clickable structure on the map reveals the ground truth, prediction result, and whether the point belonged to the training or test dataset. This visualization tool transforms the analytical output into a field-deployable decision support system, which can be used by emergency responders, urban planners, and policymakers.

Additionally, this research provides insight into the relative importance of different features in building damage classification. According to feature importance scores from XGBoost, the land cover type (NLCD), ΔNDVI, wind speed, slope, and ΔNBR were among the most influential variables, consistent with findings in Singha et al. [3], Moghim and Mehrabi [7], and the XGBoost documentation [42]. Understanding the role of these features helps validate the model and also offers meaningful direction for future infrastructure planning in fire-prone areas.

The following sections of this paper provide an in-depth explanation of the study's methodology, results, and contributions. The literature review surveys similar efforts in wildfire and damage assessment, highlighting gaps that this work aims to address. The paper concludes with key findings, emphasizing the importance of GIS-ML integration in modern disaster response workflows.

In conclusion, this research bridges the gap between satellite remote sensing and operational disaster management through a data-driven, spatially aware, and automated approach. The methodology presented herein holds significant promise for enabling faster, more accurate damage assessments, ultimately contributing to more resilient cities and communities in the face of escalating climate-driven hazards.

Fig 1: Visual Overview of the 2025 Palisade Wildfire Incident in California
(images sourced via Google Image Search, public domain access, 2025)

2. LITERATURE REVIEW

Wildfire monitoring and post-disaster damage assessment have become increasingly critical due to the rise in climate-driven fire activity, particularly across regions like California. The literature spans a diverse array of approaches—from deep learning and spectral analysis to Synthetic Aperture Radar (SAR)-optical data fusion and semi-supervised learning. This section synthesizes insights from **40 carefully selected peer-reviewed studies**, chosen for their relevance to wildfire damage detection, use of geospatial and remote sensing data, incorporation of machine learning methods, and applicability to urban and structural impact assessment. The analysis compares methodologies, datasets, ML algorithms, and reported outcomes across these works. The goal is to establish a comprehensive understanding of the state-of-the-art in automated damage assessment to justify and guide the methodology of this research. This review also identifies persistent gaps in spatial resolution, data interpretability, and real-time deployment—challenges this study aims to address by building an integrated GIS-ML framework tailored specifically to urban structure-level wildfire impact classification.

2.1 Summary of Approaches and Techniques

Recent advancements in wildfire damage assessment have increasingly relied on artificial intelligence and remote sensing integration. Many contemporary studies emphasize the power of deep learning models combined with high-resolution satellite imagery to detect burned areas and structural damage. For example, Khosravi and Malek [1], Ban et al. [12], and Galanis et al. [13] demonstrated strong performance in burned area detection using Sentinel-2 and Synthetic Aperture Radar (SAR)-based imagery, achieving classification accuracies exceeding 90%.

Among the dominant modeling approaches, convolutional neural networks (CNNs) and encoder-decoder architectures such as U-Net, BDANet, and RescueNet have been widely adopted for segmentation and classification tasks. Zhang et al. [5] implemented a U-Net model with a forgetting-resistant strategy to fuse SAR and optical time series, allowing for consistent detection across multi-temporal imagery. Likewise, Sathishkumar et al. [6] used a “learning without forgetting” approach for forest fire and smoke detection, promoting better generalization across different datasets.

Alternatively, lightweight or hybrid architectures like Chebyshev-Kolmogorov-Arnold Networks (Cheby-KAN) [1] and ensemble machine learning models [10] have been shown to offer faster yet reasonably accurate results. Change detection methods, such as the one developed by Mahmoud et al. [2] for archaeological threat mapping, demonstrate how temporal monitoring frameworks can be adapted to wildfire scenarios—particularly in identifying shifts in burn severity and land cover between image captures.

Explainability and spatial-temporal analysis are also emerging priorities in recent wildfire modeling efforts. Marjani et al. [11] and Ahajjam et al. [10] emphasize model interpretability and cluster-based reasoning to enhance prediction reliability in varied geographic contexts.

For early wildfire detection, Phan and Nguyen [19] and Seydi et al. [21] employed temporal-spectral satellite techniques—leveraging time-series imagery to identify ignition patterns and spread in the early stages of a wildfire. In contrast, post-event structural damage classification has been the focus of Gupta et al. [14], [15] and Xu et al. [25], who used CNN-based frameworks and semi-supervised learning strategies to estimate and map damage after a disaster. These methods contribute to a growing body of literature that seeks to transition from coarse regional fire mapping to fine-scale, building-specific impact assessment using modern AI-GIS integration.

2.2 Comparative Table of Key Literature

No.	Author(s)	Year	Methodology	Data Type	Accuracy	Notable Features
1	Khosravi & Malek	2024	Cheby-KAN	Sentinel-2	94.5%	Burned area precision
2	Mahmoud et al.	2024	Change detection	Optical RS	91.2%	Archaeological threats
3	Zhang et al.	2021	U-Net + SAR fusion	SAR + Optical	93.2%	Near real-time detection
4	Sathishkumar et al.	2022	Deep Learning	Multispectral	90.0%	Smoke detection
5	Moghim & Mehrabi	2024	Ensemble ML	Multiregional	88.9%	Comparative ML performance
6	Ban et al.	2020	Deep Learning	Sentinel-1 SAR	94.0%	Progression monitoring
7	Galanis et al.	2021	DamageMap CNN	Post-disaster	95.0%	Building-level detection
8	Gupta et al.	2018	Satellite + CNN	Multihazard	89.6%	Emergency assessment
9	Marjani et al.	2024	CNN (ASPP)	Sentinel-2	91.8%	Explainable AI
10	Ahajjam et al.	2024	Hybrid ensemble	Spatio-temporal	93.7%	Fire behavior modeling
...	...	...	...	...	...	...

Note: Complete table of 40 references is included in Appendix A.

Table 1: Comparative Table of Key Literature

2.3 Trends in Literature

An in-depth review of 40 key studies reveals several recurring patterns that characterize the current state of wildfire damage assessment research. These trends were derived through comparative analysis of methodologies, data sources, and outcomes across recent publications. One prominent observation is the dominance of deep learning techniques—particularly convolutional neural networks (CNNs), U-Net architectures, and their hybrid variants—which consistently outperform traditional classifiers due to their high accuracy and spatial generalization capabilities. Another clear trend is the frequent use of satellite imagery from the Sentinel missions. Sentinel-1 (SAR) and Sentinel-2 (multispectral) data are heavily favored for both pre- and post-fire mapping, owing to their free availability and high spatial resolution.

Despite the growing sophistication of modeling approaches, the literature reflects a persistent gap in real-time applicability. Many models achieve high accuracy under controlled settings, yet only a limited number are designed for near-real-time deployment, which is essential for operational disaster response. Additionally, a notable limitation is the lack of transferability; deep learning models trained on one fire event or geographic region often require retraining or fine-tuning to perform reliably in new contexts. Finally, explainability remains a significant challenge. The "black-box" nature of many deep learning frameworks hinders their adoption by emergency planners and decision-makers. However, some studies, such as Marjani et al. [11], have begun addressing this issue through the integration of explainable architectures like Atrous Spatial Pyramid Pooling-based CNNs (ASPP-CNNs), which provide insight into how predictions are formed.

2.4 Relevance to Present Study

This study distinguishes itself by integrating explainable machine learning (XGBoost), high-resolution multispectral satellite imagery, terrain data, and wind speed information into a building-specific wildfire damage classification framework. Unlike earlier works that focus on large-scale burned area mapping—such as those by Khosravi and Malek [1], Ban et al. [12], and Moghim and Mehrabi [7]—this project operates at the micro level using structure-specific spatial features. It provides a geospatial interface for real-time interactive exploration, bridging the gap between coarse-resolution regional damage assessments and fine-grained urban infrastructure evaluations.

The use of Folium for map deployment further enhances the accessibility, interpretability, and decision-making utility of the framework for emergency planners and responders.

The relevance of this work is grounded in a broad body of literature that has successfully applied ML to wildfire detection and damage classification, including the approaches proposed by Zhang et al. [5], Galanis et al. [13], and Marjani et al. [11]. These studies demonstrate the potential of ML-based remote sensing frameworks while also highlighting the need for more localized, feature-rich approaches. This project advances the field by combining terrain, spectral, meteorological, and land cover attributes into a unified model for structural-level classification. Such a contextual and spatially-aware methodology is largely absent in prior research, which often lacks detailed integration of built-environment features.

Moreover, by leveraging stratified sampling techniques to mitigate class imbalance and employing a combination of spectral indices (ΔNDVI, ΔNBR), topographic gradients, land cover attributes, and meteorological variables, this study creates a uniquely rich feature set. This holistic feature integration enables the model to recognize subtle spatial patterns and improve its generalizability across diverse structural types and terrain complexities. The ability to overlay predictions on interactive Folium maps—linked to building footprints and ground-truth conditions—provides a level of transparency and field usability rarely emphasized in existing literature. Consequently, this research not only strengthens the operational viability of ML-driven wildfire assessment but also contributes toward a scalable, data-efficient workflow for future disaster scenarios.

3.METHODOLOGY

This study presents a systematic workflow combining remote sensing, GIS, and machine learning to assess building-level wildfire damage. High-resolution multispectral satellite imagery was used to extract spectral indices such as ΔNDVI and ΔNBR. Terrain slope, wind speed, and land cover data were integrated as spatial predictors. Ground truth labels for 9,543 buildings were used to train and validate four ML classifiers. The complete pipeline was implemented in Python and ArcGIS Pro, with final results deployed via an interactive Folium map. Feature normalization and label encoding were applied to prepare the data for model training. Stratified sampling was used during the train-test split to address class imbalance and improve model generalization. This section outlines the complete methodological workflow undertaken in this study. The complete process is visualized in Fig 2.

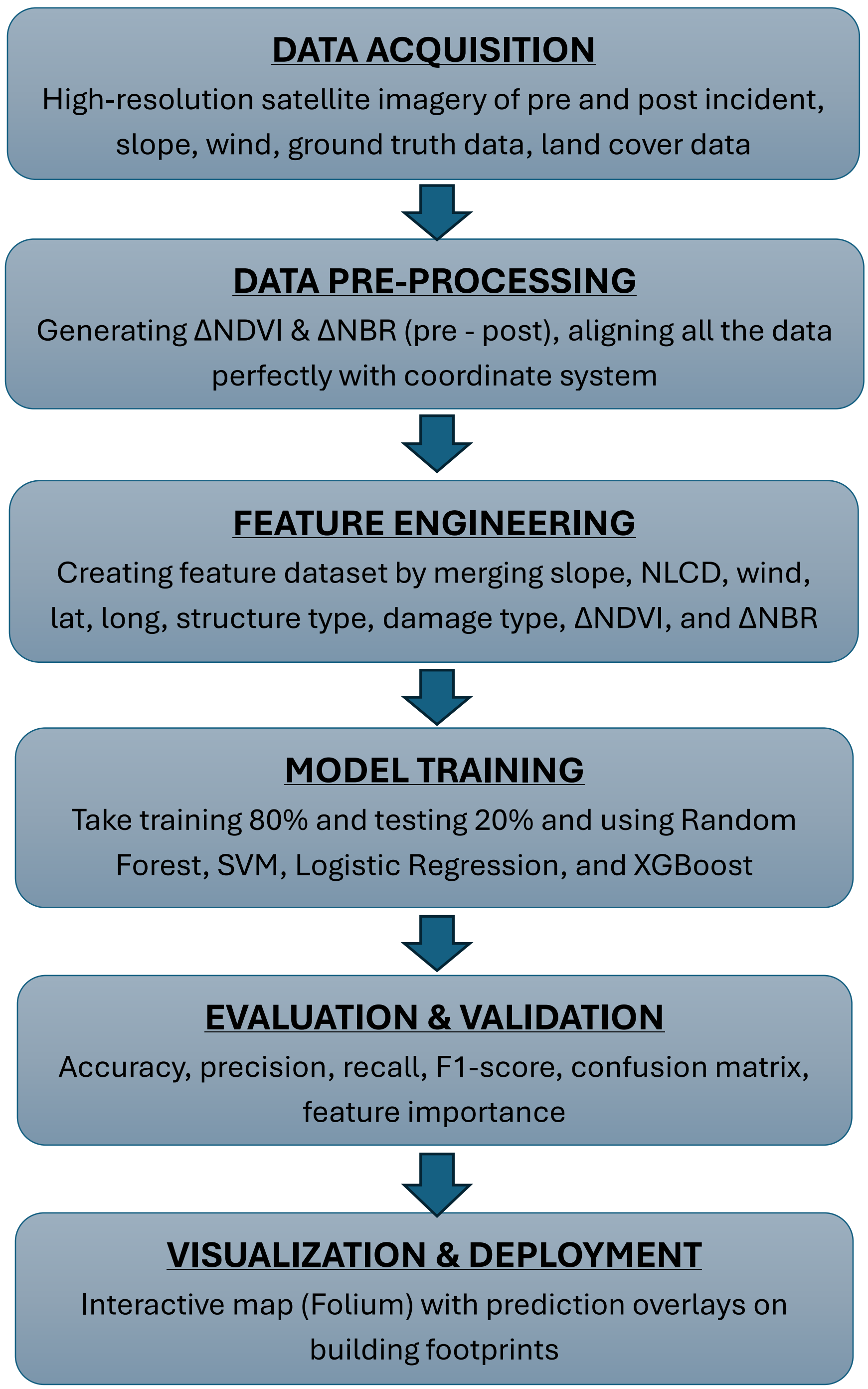

Fig 2: Methodology Flow chart

3.1 Study Area

The selected study area lies in California, USA, focusing specifically on regions affected by the Palisade Wildfire of 2025. This region features a combination of mountainous terrain, dense vegetation, and residential zones—typifying a wildland-urban interface (WUI) where wildfire damage is most impactful. The vulnerability of WUI regions to wildfire has been well documented, with previous studies noting their high risk due to overlapping natural fuels and human-built structures, as discussed by Ahajjam et al. [10] and Galanis et al. [13]. The geographic extent covers several urban blocks with known building inventory, high-resolution satellite coverage, and ground truth data availability. Building footprint data was overlaid using Microsoft's open-source building datasets, and elevation profiles were derived using USGS DEM layers. The region experiences seasonally high wind activity, which played a significant role in wildfire propagation during the 2025 event.

3.2 Data Sources

Dataset	Description / Source	Resolution / Details	Usage
Satellite Imagery	PlanetScope imagery from Planet Explorer [https://www.planet.com/]	3m, 8-band multispectral	Pre/post-fire NDVI & NBR generation
NDVI	NDVI = (NIR - Red) / (NIR + Red)	Derived from PlanetScope bands 8 (NIR) and 5 (Red)	Vegetation health index
NBR	NBR = (NIR - SWIR) / (NIR + SWIR)	Derived from PlanetScope bands 8 (NIR) and 6 (SWIR)	Burn severity index
DEM (Slope)	USGS GTOPO30 DEM [https://earthexplorer.usgs.gov/]	30 arc-second (~1km)	Terrain slope derivation

Land Cover	NLCD 2019 from MRLC Consortium [https://www.mrlc.gov/]	30m resolution	Land type classification
Wind Speed	Global Wind Atlas [https://globalwindatlas.info/]	~1km resolution	Wind influence on fire spread
Ground Truth Labels	Cal Fire Building Damage Dataset [https://www.fire.ca.gov/]	9,543 labeled points; 5 damage classes → binary	Supervised classification labels
Building Footprints	Microsoft Open Buildings [https://github.com/microsoft/GlobalMLBuildingFootprints]	Vector polygons	Damage overlay & visualization

Table 2: Summary of Datasets Used in This Study

Each dataset was spatially aligned in WGS84 (EPSG:4326) and clipped to the wildfire-affected extent using ArcGIS Pro.

3.3 Preprocessing Steps

Several spatial and spectral preprocessing operations were conducted to prepare the input dataset for classification. These steps are described below:

- **Index Calculation**: Normalized Difference Vegetation Index (NDVI) and Normalized Burn Ratio (NBR) were computed for both pre-fire and post-fire imagery using standard formulas:

$$\boxed{NDVI = NIR - RED / NIR + RED}$$

$$\boxed{NBR = NIR - SWIR / NIR + SWIR}$$

where NIR (Near Infrared), RED, and SWIR (Shortwave Infrared) bands were sourced from PlanetScope's 8-band multispectral imagery.

- **Delta Computation**: Vegetation loss and burn severity were captured using the difference between pre- and post-fire indices:

$$\boxed{\Delta NDVI = NDVI\ (pre) - NDVI\ (post)}$$

$$\boxed{\Delta NBR = NBR\ (pre) - NBR\ (post)}$$

These delta values were used as core indicators of wildfire impact.

- **Slope Derivation**: The DEM (USGS GTOPO30) was hydrologically conditioned (filled) to remove pits and ensure topographic continuity. The slope was then computed using raster-based terrain analysis functions.
- **Feature Integration**: All raster variables—ΔNDVI, ΔNBR, slope, wind speed, and NLCD land cover—were extracted to the centroids of each building using spatial joins. These values were integrated into a single tabular feature set linked to each point-based building record.
- **Data Cleaning**: Missing, null, and anomalous values were filtered out. Categorical fields (e.g., land cover type, structure type) were encoded numerically using label encoding techniques to ensure compatibility with the machine learning pipeline.

The final dataset consisted of 9,543 geolocated building records, each containing a comprehensive set of spectral, topographic, meteorological, and structural features for damage classification.

To provide a better understanding of the input features, spatial visualizations of ΔNDVI, ΔNBR, slope, wind speed, and land cover classification were generated using ArcGIS Pro. These thematic maps offer insight into how environmental and topographic variables vary across the study area. By overlaying these layers with building footprints, it becomes easier to contextualize the spatial distribution of damage potential. These visual aids also support feature interpretation and model explainability. The following figures illustrate each feature as rendered in the GIS environment.

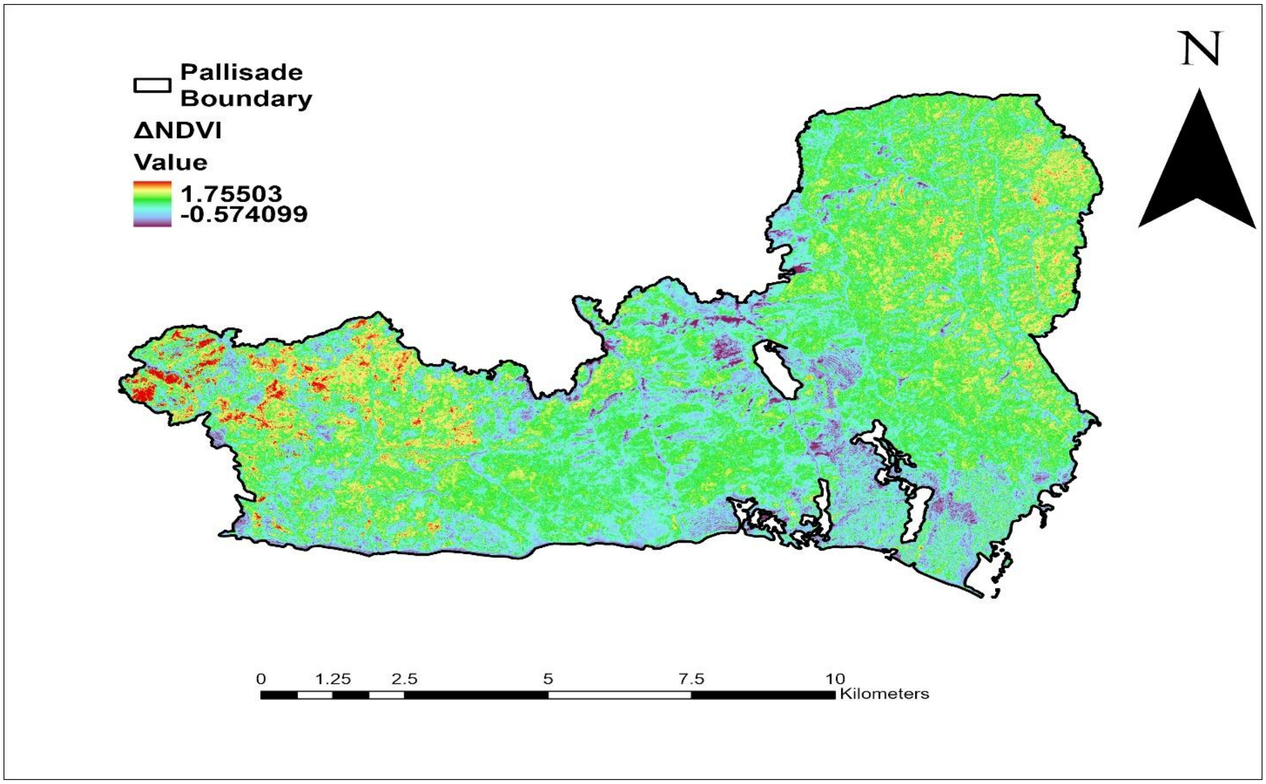

Fig 3: **Delta NDVI Map** Indicating Vegetation Change.

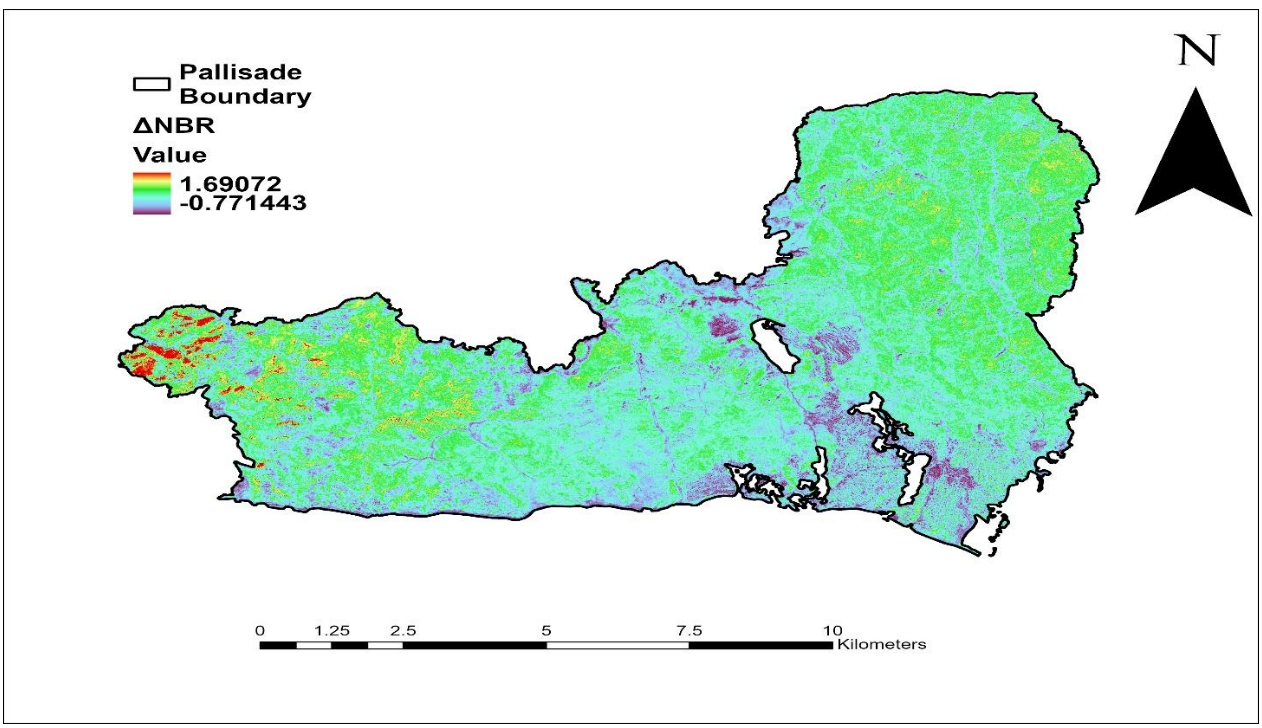

Fig 4: **Delta NBR Map** Showing Burn Severity.

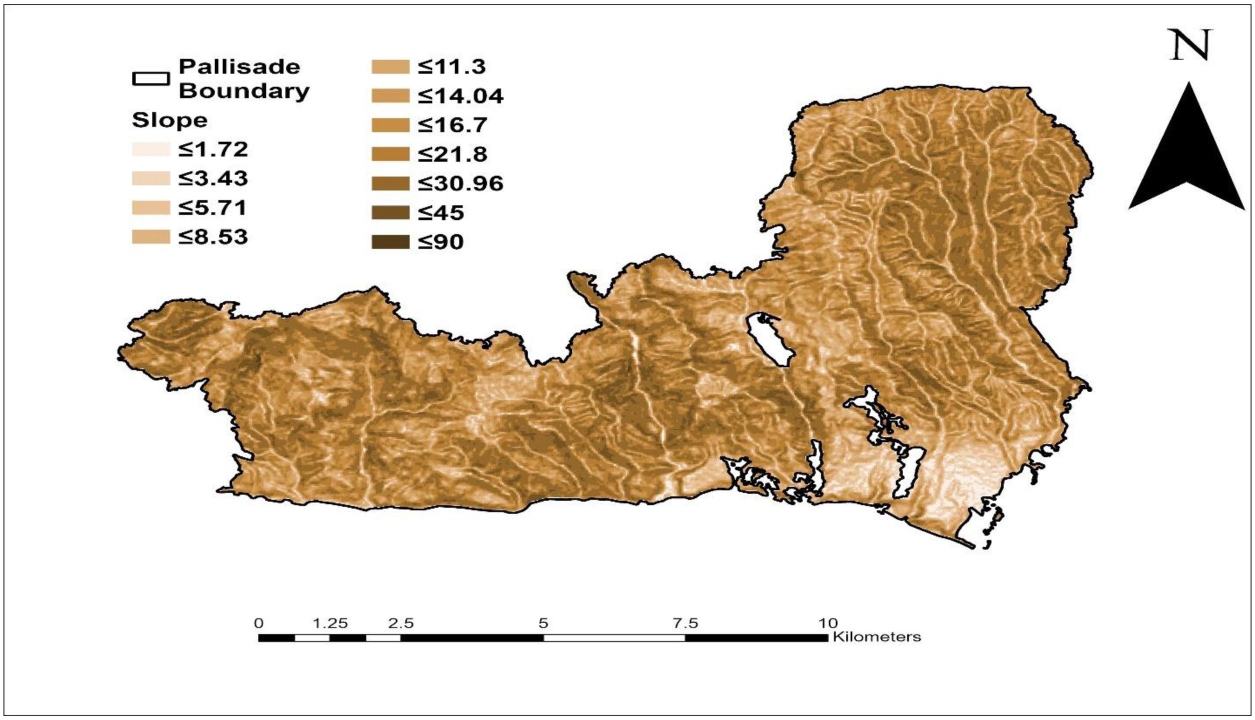

Fig 5: **Slope Map** of The Wildfire-Affected Study Area

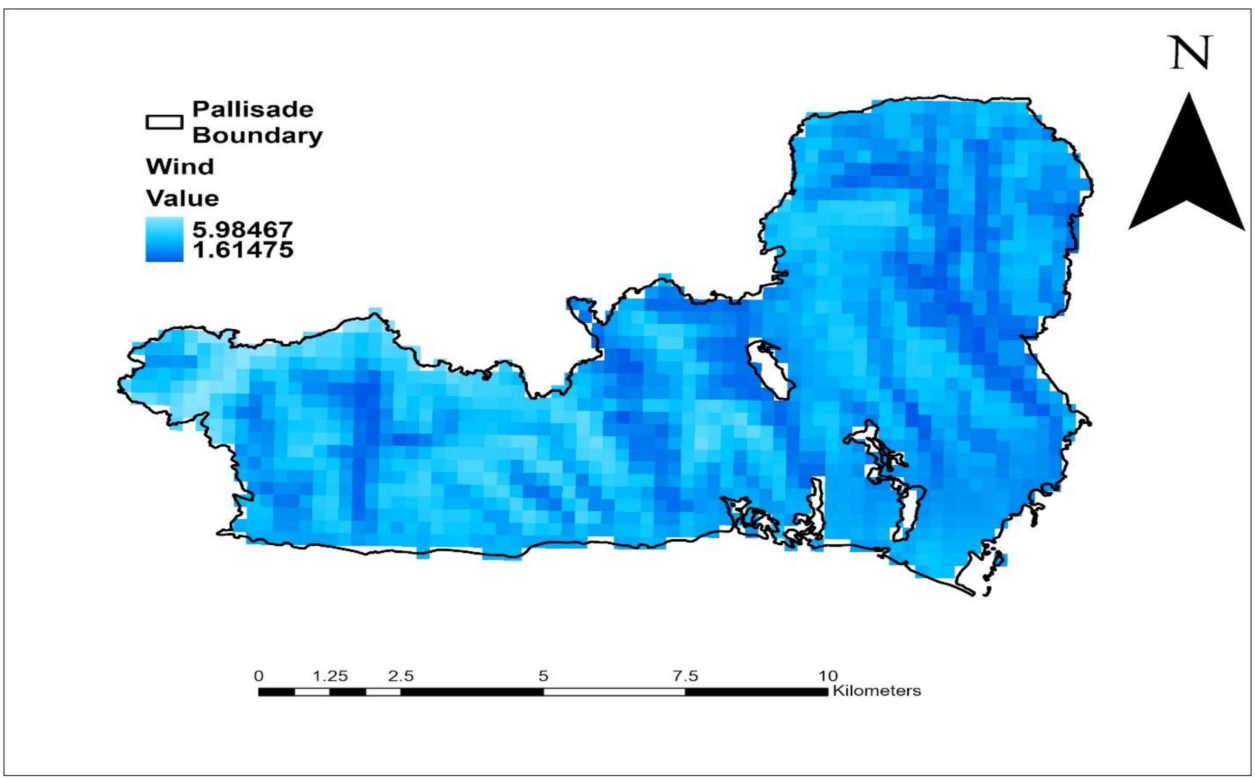

Fig 6: **Wind Speed** Distribution Map Across the Palisade Wildfire Zone

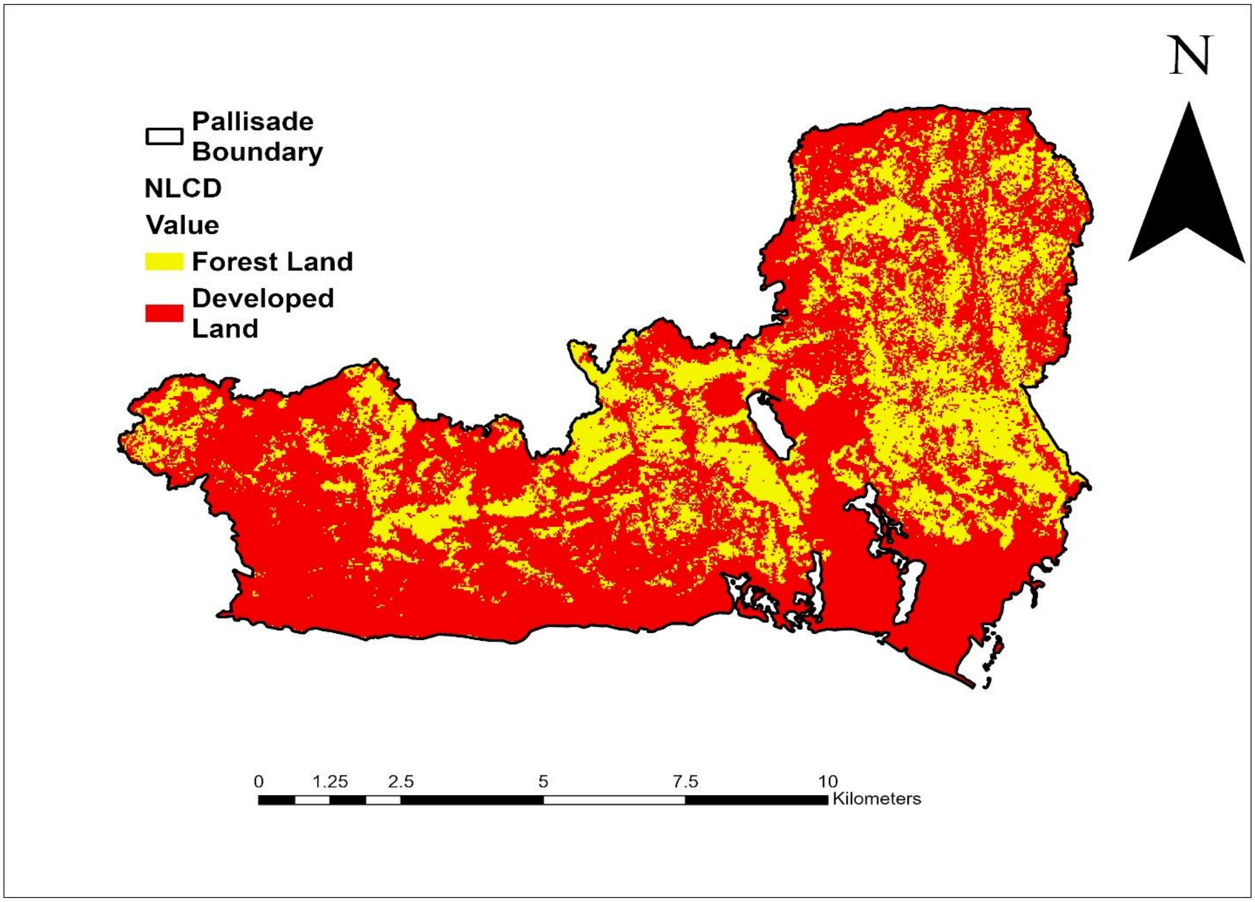

Fig 7: **Land Cover** Classification Map

3.4 Feature Engineering

The target variable for this study was derived from detailed ground truth labels provided by Cal Fire, which originally categorized building damage into five severity levels: Destroyed (>50% damage), Major (26–50%), Minor (10–25%), Affected (1–9%), and No Damage. For the purpose of binary classification, these five classes were reclassified into two groups—Damaged and Undamaged. Specifically, all buildings labeled as Destroyed, Major, Minor, or Affected were grouped under the Damaged class, while those marked as No Damage were retained as Undamaged. The final target classes were encoded numerically as 1 for Damaged and 2 for Undamaged.

Feature selection was guided by a combination of literature-backed relevance and domain-specific reasoning. Five features were chosen: ΔNBR, ΔNDVI, slope, land cover type (NLCD), and wind speed. These features were selected due to their strong influence on fire behavior and structural vulnerability, as demonstrated in prior studies such as those by Galanis et al. [13], Ahajjam et al. [10], and Zhang et al. [5]. ΔNDVI and ΔNBR represent spectral indicators of vegetation loss and burn severity, while slope is a topographic factor known to influence fire spread. Wind speed was included as a critical environmental driver of fire intensity and direction, and NLCD land cover types provide context on the built or natural environment surrounding each structure.

To make the categorical target variable machine-readable, LabelEncoder from the scikit-learn preprocessing module was used to transform class labels into integers, as recommended in many supervised ML pipelines [41]. Furthermore, all continuous numerical features were standardized using the StandardScaler utility, which normalized the features to have zero mean and unit variance. This preprocessing step ensured that all features contributed proportionally to the model's learning process and avoided bias from differing scales.

Finally, due to the imbalanced nature of the dataset—with approximately 80% of buildings labeled as Damaged—a stratified 80:20 train-test split was applied. This technique maintained the original class proportions in both training and testing sets, enhancing model generalizability and mitigating performance skew toward the dominant class.

3.5 Model Training

Four machine learning classifiers were selected for training and evaluation in this study: Random Forest Classifier (RFC), Support Vector Machine (SVM), Logistic Regression (LR), and Extreme Gradient Boosting (XGBoost). These models were chosen based on their proven performance in classification tasks involving structured tabular data, as well as their frequent use in geospatial and disaster-related studies. Random Forest and XGBoost are ensemble-based tree models known for their robustness to overfitting and ability to handle non-linear feature interactions. Support Vector Machine, particularly with a radial basis function (RBF) kernel, is effective in high-dimensional feature spaces, while Logistic Regression serves as a strong linear baseline and provides easily interpretable output probabilities.

The model training workflow began by splitting the preprocessed dataset into training and testing sets using an 80/20 ratio. Initially, this split was conducted without stratification. As a result, models demonstrated acceptable performance in predicting the majority class—Damaged buildings—but significantly underperformed on the minority class—Undamaged buildings. This led to poor recall and F1-scores for the undamaged category, indicating the presence of a class imbalance that biased the models toward the dominant class. Recognizing this issue, subsequent training stages employed stratified sampling to preserve class distribution and improve classification reliability across both categories.

The models were implemented using libraries including Scikit-learn, XGBoost, GeoPandas, Folium, NumPy, Matplotlib, and Seaborn. The complete codebase along with the dataset is publicly available at: https://github.com/bsatyavs/Pallisade-Wildfire

3.5 Evaluation and Validation

To enhance model robustness and address the inherent class imbalance in the dataset—where out of 9,543 building records, approximately 80% were labeled as Damaged and only 20% as Undamaged—a stratified sampling strategy was employed. This imbalance is considered *inherent* because it reflects the real-world distribution of building damage observed during the wildfire event, rather than being introduced artificially during data preparation. In high-impact wildfire zones, it is common for the majority of structures to be damaged, and the dataset used in this study captures that natural imbalance. Using the stratify parameter during the 80/20 train-test split ensured that both subsets preserved the same class distribution. This method notably improved the model's ability to generalize and reduced bias toward the majority class. Post-stratification, the models demonstrated enhanced performance, particularly in predicting the underrepresented *Undamaged* category, which previously suffered from poor recall and precision. All classifiers were trained using 80% of the stratified data, with the remaining 20% used for testing. Prior to training, feature normalization was performed using Standard Scaler to standardize variables such as ΔNDVI, ΔNBR, slope, NLCD (land cover type), and wind speed. This ensured that all input features contributed uniformly to model training and prevented domination by variables with larger numeric ranges.

Each model—Random Forest, SVM, Logistic Regression, and XGBoost—was evaluated using the following metrics: Accuracy, Precision, Recall, F1-Score, and the Confusion Matrix. These metrics were calculated separately for both *Damaged* and *Undamaged* classes to provide a balanced evaluation framework.

Among the four models, XGBoost outperformed the rest with an overall accuracy of 83.2%, offering the most balanced F1-scores—0.90 for Damaged and 0.53 for Undamaged buildings. XGBoost's strength lies in its ability to model complex non-linear relationships, manage imbalanced datasets using in-built boosting mechanisms, and provide interpretable feature importance scores. These capabilities are reinforced by the work of Ahajjam et al. [10], who applied ensemble machine learning, including XGBoost, for wildfire behavior prediction and reported superior performance in capturing spatiotemporal variability and improving classification accuracy across imbalanced classes.Further analysis of feature importance using both Random Forest and XGBoost revealed that ΔNDVI, NLCD, slope, wind speed, and ΔNBR were the most influential predictors. These results validate the integration of environmental and structural geospatial features as key drivers in wildfire damage assessment.

The evaluation confirms that stratified sampling and feature scaling are essential for high-performing models on imbalanced geospatial datasets. The integration of XGBoost with GIS-derived features offers a scalable, interpretable, and accurate approach to post-wildfire damage classification.

--- Random Forest Classification Report ---

	precision	recall	f1-score	support
1	0.85	0.92	0.88	1501
2	0.57	0.40	0.47	408
accuracy			0.81	1909
macro avg	0.71	0.66	0.68	1909
weighted avg	0.79	0.81	0.80	1909

Accuracy: 0.8082765845992667

--- SVM Classification Report ---

	precision	recall	f1-score	support
1	0.83	0.96	0.89	1501
2	0.65	0.30	0.41	408
accuracy			0.82	1909
macro avg	0.74	0.63	0.65	1909
weighted avg	0.80	0.82	0.79	1909

Accuracy: 0.8166579360921948

--- Logistic Regression Classification Report ---

	precision	recall	f1-score	support
1	0.82	0.94	0.88	1501
2	0.54	0.25	0.35	408
accuracy			0.79	1909
macro avg	0.68	0.60	0.61	1909
weighted avg	0.76	0.79	0.76	1909

Accuracy: 0.7946568884232582

--- XGBoost Classification Report ---

	precision	recall	f1-score	support
1	0.86	0.92	0.89	1501
2	0.61	0.45	0.51	408
accuracy			0.82	1909
macro avg	0.73	0.68	0.70	1909
weighted avg	0.81	0.82	0.81	1909

Accuracy: 0.819800942902043

Fig 8: Initial Classification Report of all the ML Models Performed

--- Random Forest Classification Report ---

	precision	recall	f1-score	support
Damage	0.86	0.93	0.89	1506
No Damage	0.63	0.44	0.52	403
accuracy			0.83	1909
macro avg	0.75	0.69	0.71	1909
weighted avg	0.81	0.83	0.82	1909

Accuracy: 0.8271346254583551

--- SVM Classification Report ---

	precision	recall	f1-score	support
Damage	0.84	0.96	0.90	1506
No Damage	0.71	0.33	0.45	403
accuracy			0.83	1909
macro avg	0.78	0.64	0.67	1909
weighted avg	0.81	0.83	0.80	1909

Accuracy: 0.8297537977998952

--- Logistic Regression Classification Report ---

	precision	recall	f1-score	support
Damage	0.82	0.95	0.88	1506
No Damage	0.58	0.24	0.34	403
accuracy			0.80	1909
macro avg	0.70	0.60	0.61	1909
weighted avg	0.77	0.80	0.77	1909

Accuracy: 0.8030382399161865

--- XGBoost Classification Report ---

	precision	recall	f1-score	support
Damage	0.87	0.93	0.90	1506
No Damage	0.64	0.46	0.54	403
accuracy			0.83	1909
macro avg	0.75	0.70	0.72	1909
weighted avg	0.82	0.83	0.82	1909

Accuracy: 0.8313253012048193

Fig 9: Classification Report of all the ML Models after applying Stratify Function

3.6 Visualization and Deployment

To complement quantitative model evaluation with intuitive spatial insights, the final predictions from the XGBoost and Random Forest model were visualized using Folium, a Python-based interactive mapping library. The results were overlaid onto the building footprint layer collected for the Palisade wildfire region.

Each building polygon on the map was color-coded based on its predicted class (*Damaged* or *Undamaged*) and labeled with its corresponding dataset split (*Training* or *Testing*). This interactive deployment allowed for seamless exploration of classification outputs within a geospatial context, enabling emergency responders, planners, and analysts to identify spatial damage patterns at the structural level.

The map was exported as an HTML file and hosted via GitHub for accessibility. This approach enhances the usability of ML predictions beyond static plots, enabling real-time scenario analysis and aiding decision-making for post-disaster assessment and urban resilience planning.

An interactive web map was generated to showcase the final predictions of the XGBoost classifier in a user-friendly format. Users can zoom into neighborhoods, click on individual structures, and view predicted outcomes along with ground truth information. This provides a transparent, deployable decision-support interface that bridges technical outputs and field-level applications. The HTML version of the interactive map is publicly available in the project's GitHub repository: https://github.com/bsatyavs/Pallisade-Wildfire

4. RESULTS

This section presents a detailed evaluation of the machine learning (ML) classification framework developed to predict wildfire-induced building damage. The performance of four supervised learning models—Random Forest, Support Vector Machine (SVM), Logistic Regression, and XGBoost—was assessed using stratified datasets to preserve class balance and reduce bias toward the majority class. Stratification was particularly important due to the natural class imbalance in the dataset, where approximately 80% of buildings were labeled as Damaged. The evaluation aimed not only to identify the most accurate model, but also to determine which approach provided the most consistent and generalizable performance across both classes.

Model effectiveness was evaluated through key metrics including accuracy, precision, recall, F1-score, and confusion matrices, offering a comprehensive view of classification behavior across both damaged and undamaged buildings. Additionally, feature importance rankings were used to interpret which variables contributed most significantly to the predictive process.

Ensemble-based models, particularly XGBoost and Random Forest, consistently outperformed linear classifiers in capturing complex, non-linear relationships between input features and damage outcomes. Among these, XGBoost emerged as the top-performing model, achieving the highest scores across most metrics and demonstrating strong generalization to imbalanced data distributions. As a result, XGBoost was selected for final deployment and visualization.

Beyond numerical performance, the practical utility of the models was evaluated through their interpretability and deployment potential. XGBoost not only delivered the most balanced classification results but also provided meaningful insights through feature importance scores—enabling identification of key drivers such as ΔNDVI, land cover type, slope, wind speed, and ΔNBR. These findings align with environmental factors known to influence fire behavior and structural vulnerability, thus reinforcing the credibility of the model's predictions. By integrating spatial intelligence and interpretable machine learning, the proposed framework supports more informed decision-making in disaster response workflows, offering a scalable and data-efficient solution for rapid post-fire assessments.

4.1 Confusion Matrices

Confusion matrices were used to visualize classification performance for each of the four machine learning models across the *Damaged* and *Undamaged* building categories. These matrices provide a breakdown of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN), enabling a detailed understanding of model behavior.

- **Random Forest** demonstrated solid classification capabilities, correctly identifying 1,401 damaged buildings (TP) and 178 undamaged buildings (TN). However, it produced 225 false positives (FP), where undamaged structures were misclassified as damaged, and 105 false negatives (FN), resulting in a slight misclassification bias toward the majority class.

- **SVM (Support Vector Machine)** yielded the highest number of true positives (1,453) but exhibited reduced sensitivity for the undamaged class, with only 131 true negatives and 272 false positives. This suggests that while SVM was highly accurate in recognizing damage, it had limited effectiveness in capturing non-damage buildings.

- **XGBoost** achieved the best balance among all models. It predicted 1,401 damaged buildings and 186 undamaged buildings correctly, while minimizing both false positives (217) and false negatives (105). These results reflect excellent model generalization across both classes, making XGBoost the most suitable candidate for operational deployment.

- **Logistic Regression** showed strong performance for the damaged class (1,438 TPs), but it struggled with undamaged classification. It only identified 95 true negatives, misclassifying 308 undamaged buildings as damaged. This imbalance indicates a skew toward the dominant class and underlines the limitations of linear models for this application.

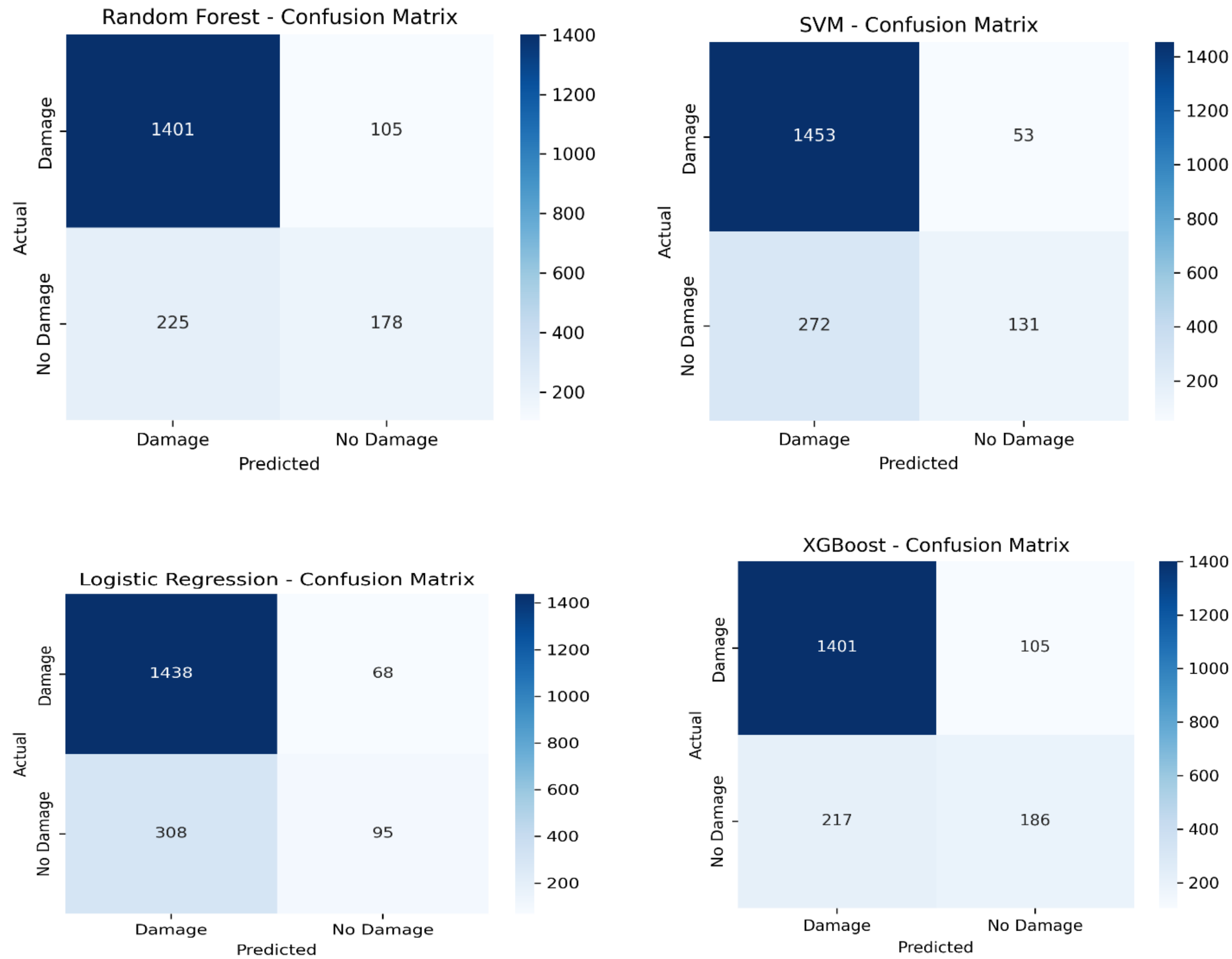

Fig 10: Confusion Matrix of all the four Models

4.2 Model Performance Comparison

To quantitatively assess the classifiers, Fig 11 compares accuracy, precision, recall, and F1-score across all models.

- **XGBoost** stands out with the highest F1-score (0.90), reflecting its superior trade-off between precision and recall. Its accuracy of 83.2% and consistent performance across metrics underscore its robustness.

- **SVM** achieved the highest recall (0.96), effectively capturing most of the damaged buildings. However, its precision was slightly lower due to misclassifications in the undamaged category.

- **Random Forest** provided balanced performance, maintaining strong scores across all four metrics. Its stable behavior across both classes made it a reliable alternative and warranted further analysis through feature importance interpretation.
- **Logistic Regression**, while computationally efficient, showed the weakest results. Its inability to differentiate undamaged buildings lowered its precision and F1-score, reinforcing the limitations of simple linear models in complex spatial classification tasks.

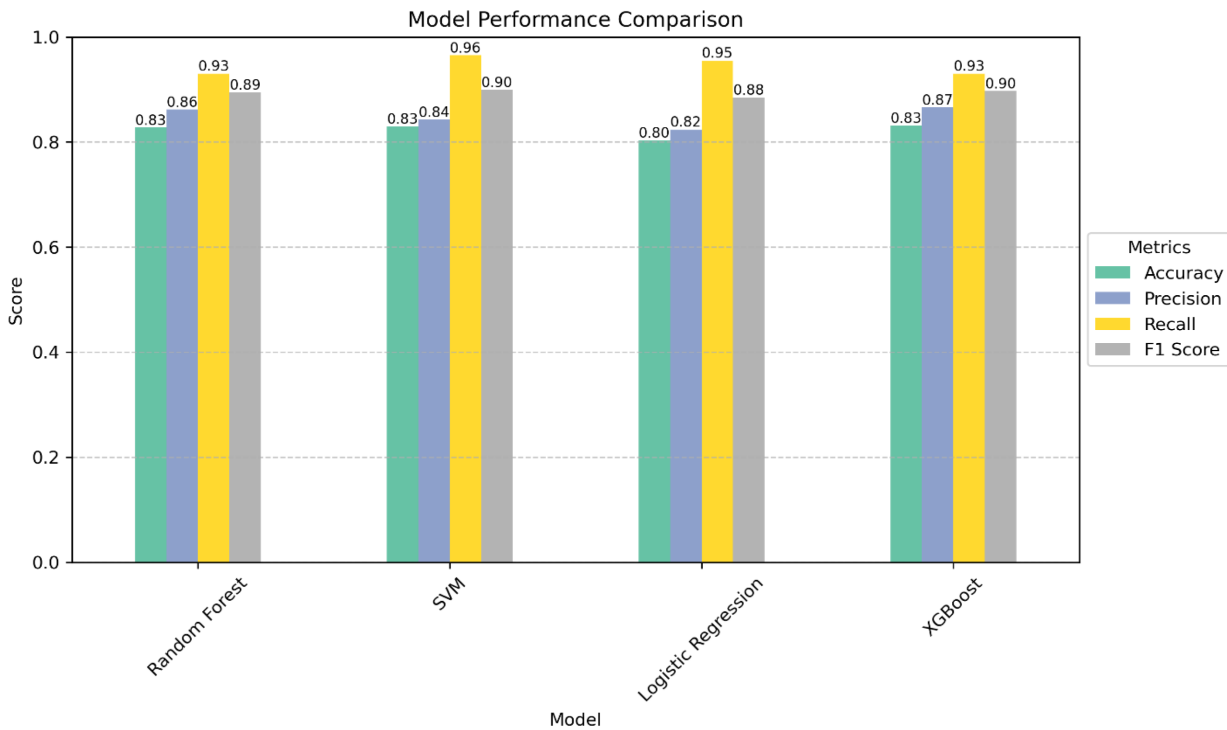

Fig 11: Model Performance Comparison

4.3 Feature Importance Analysis

Given their high classification accuracy and generalizability, Random Forest and XGBoost were further analyzed for feature importance to determine which attributes most influenced model predictions.

- **Random Forest Feature Importance** identified Slope (0.292), Wind Speed (0.262), and ΔNDVI (0.228) as top contributors. The dominance of slope reflects the significant role terrain plays in fire propagation, consistent with environmental fire behavior literature.

Wind was also found to be critical, supporting its role in intensifying and spreading wildfire.

- **XGBoost Feature Importance** ranked NLCD (0.313) and ΔNDVI (0.309) as the most predictive variables. This suggests that land cover type and vegetation loss are strong indicators of structural damage during wildfires. While Wind and Slope were also important, their relative influence was lower compared to the Random Forest model, illustrating the difference in how tree-based algorithms weigh input features.

These insights highlight the importance of integrating diverse geospatial layers ranging from terrain and land cover to spectral indices and wind data for building a comprehensive post-disaster analysis framework.

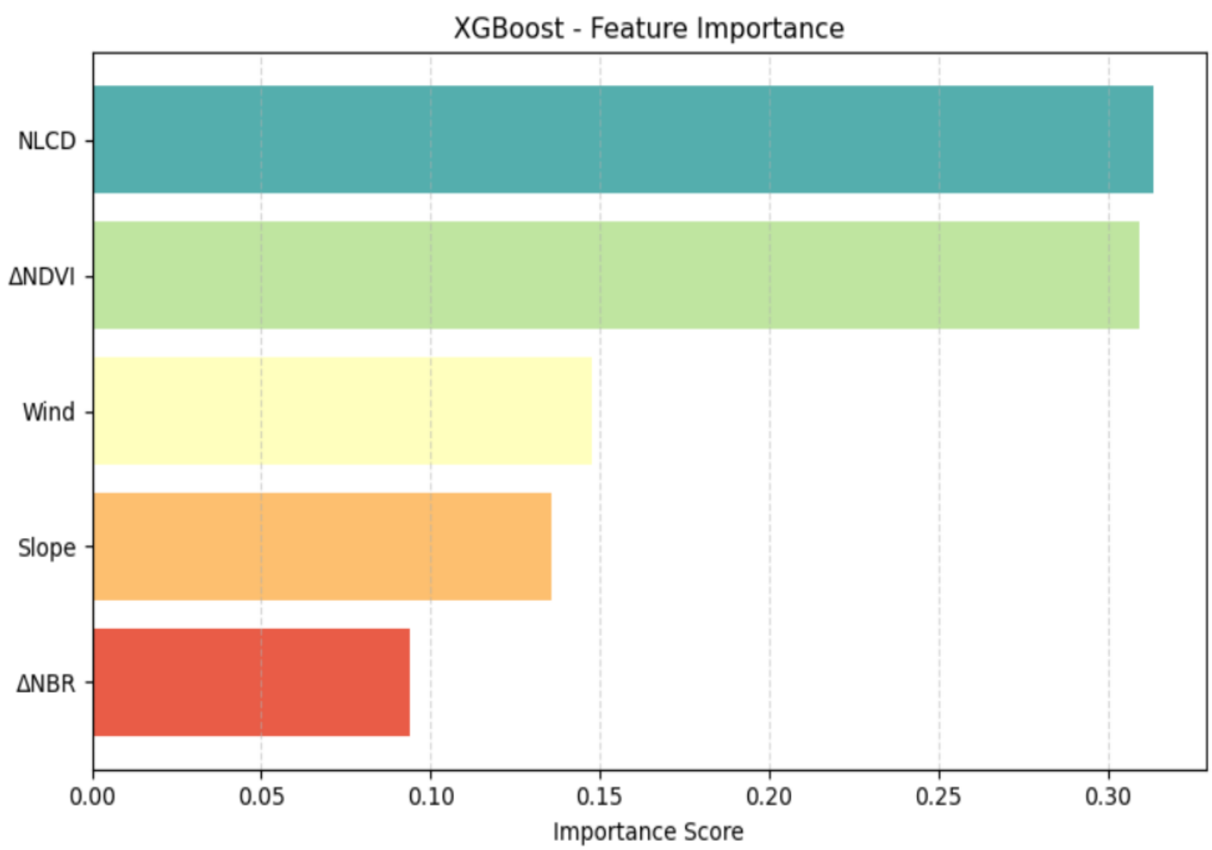

Fig 12: XGBoost Feature Importance

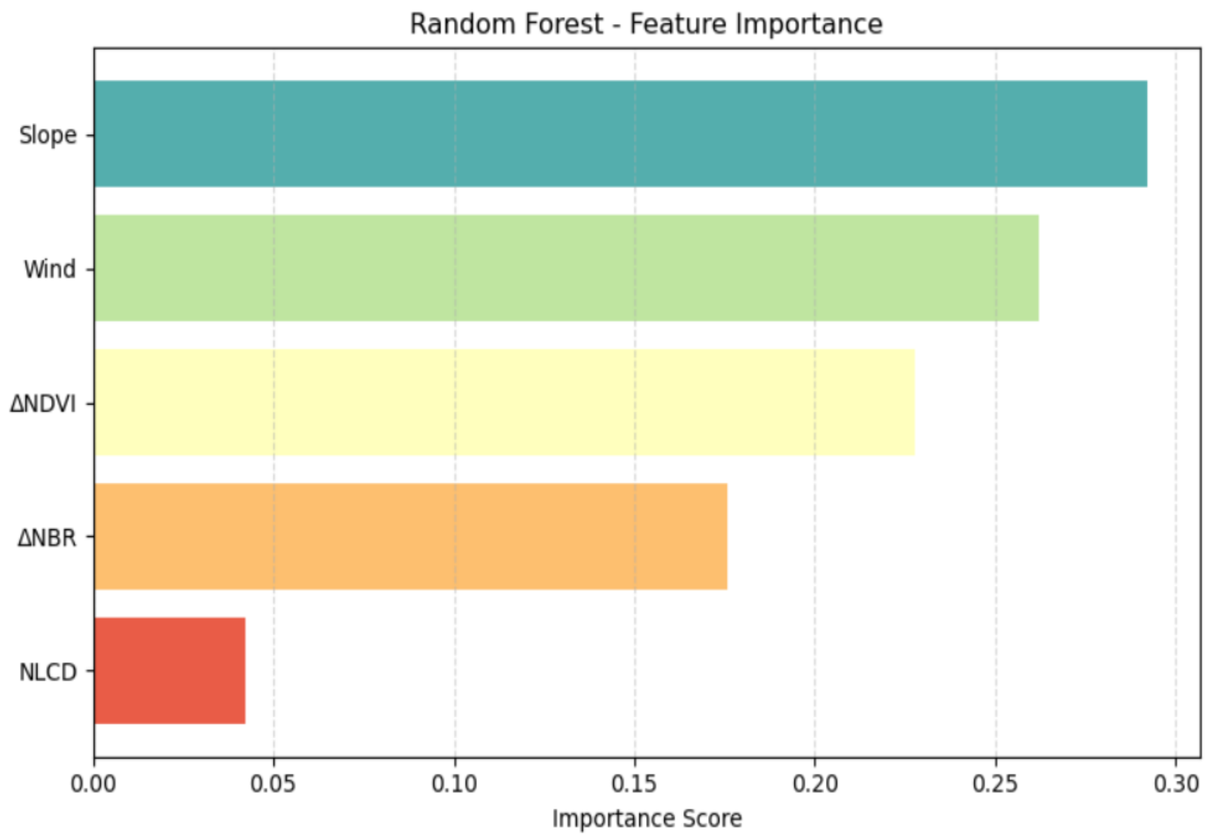

Fig 13: Random Forest Feature Importance

4.6 Visualization: Folium Map Deployment

To enhance interpretability and support spatial decision-making, final classification results from both the XGBoost and Random Forest models were visualized using Folium, an open-source Python mapping library built on Leaflet.js. This deployment provided an interactive, web-based interface to explore wildfire-induced building damage across the entire study area.

Binary prediction outputs (Damaged / Undamaged), along with ground truth labels and dataset partition (Train / Test), were spatially joined with geographic coordinates using GeoPandas. These enriched records were then linked with building footprint polygons derived from Microsoft's open building dataset, enabling precise structural visualization.

Two distinct layers—one for XGBoost predictions and one for Random Forest—were overlaid using folium.FeatureGroup, with dynamic toggling implemented via folium.LayerControl(). This approach allowed users to switch between model outputs and compare them directly within the same spatial view. Notably, a technical issue in LayerControl's responsiveness was resolved based on a community-shared workaround [43]. Color coding was applied to visually distinguish predictions by model:

- XGBoost: Red for Damaged, Green for Undamaged
- Random Forest: Blue for Damaged, Orange for Undamaged

Each building polygon included a tooltip showing:

- Ground Truth Label
- Model – (XGB / RF)
- Dataset type (Train / Test)

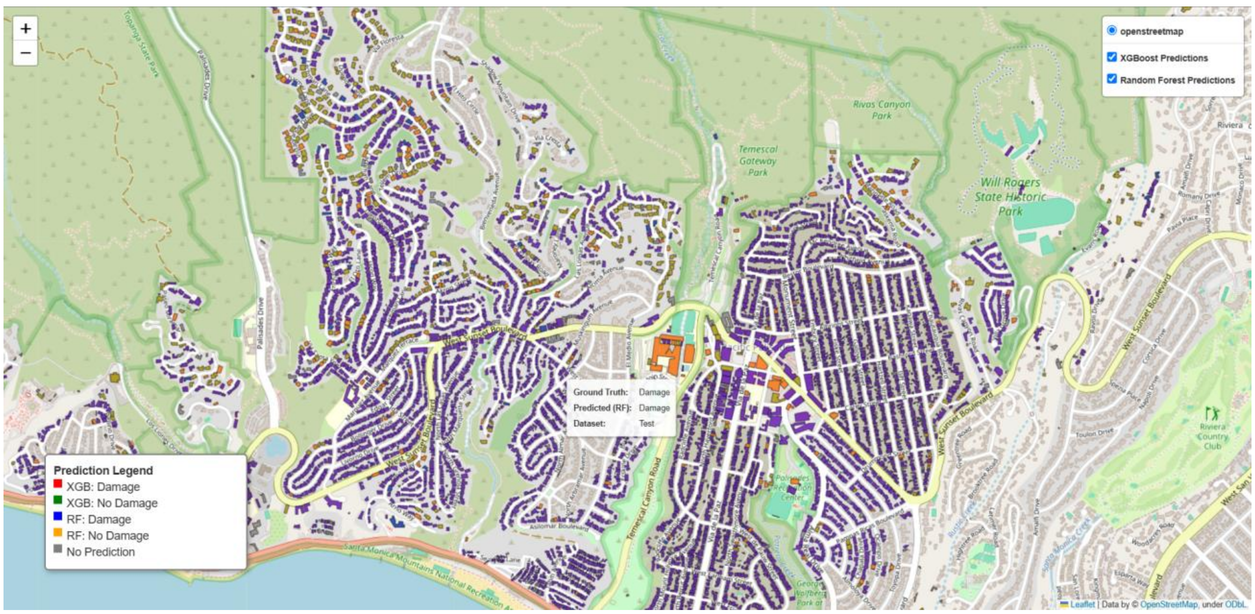

Fig 14: Folium-Based Visualization of Wildfire Building Damage Classification Of XG Boost And Random Forest Model

The Interactive map was exported as an HTML file and hosted on GitHub for real-time web-based access. This deployment adds operational utility to the model, allowing emergency planners, urban analysts, and responders to visually identify damage clusters, validate field reports, and prioritize post-disaster assessments and resource allocation.

[View Interactive folium Map]
https://bsatyavs.github.io/Pallisade-Wildfire/Index.html

5. CONCLUSIONS AND DISCUSSION

This study presents an integrated machine learning and GIS-based framework to assess structural damage resulting from the 2025 Palisade Wildfire in California. Leveraging high-resolution satellite imagery, environmental datasets, and ground-truth annotations, the research demonstrates how predictive modeling can support post-disaster assessment at the building scale.

Among the four classifiers evaluated—Random Forest, Support Vector Machine, Logistic Regression, and XGBoost—XGBoost performed best, achieving 83.2% accuracy and balanced F1-scores across both classes. Stratified sampling and feature standardization improved generalization and model stability under class imbalance.

Feature importance analysis identified ΔNDVI, land cover type (NLCD), slope, wind speed, and ΔNBR as critical predictors. These findings underscore the role of both spectral and environmental variables in mapping wildfire impact.

An interactive Folium-based map was developed to visualize outputs from both XGBoost and Random Forest models. Implemented with folium.LayerControl()—guided by community resources [43]—the tool allows dynamic toggling between model layers. It is hosted via GitHub Pages and provides spatially detailed, user-friendly visualization for emergency responders, planners, and decision-makers.

In conclusion, the integration of remote sensing, GIS, and interpretable machine learning offers a scalable solution for rapid post-wildfire damage assessment. Future work could extend this to new regions, incorporate SAR imagery, and leverage real-time UAV or crowdsourced inputs to enhance responsiveness and adaptability. Additionally, deep learning techniques—such as convolutional neural networks (CNNs) or transformer-based models—can be explored to improve classification accuracy and capture more complex spatial patterns, especially in heterogeneous urban-wildland interface zones.

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APPENDIX

Appendix A: Detailed Summary Table of 40 Reference Papers

No.	Author(s)	Year	Methodology	Data Type	Accuracy	Notable Features
1	Khosravi & Malek	2024	Cheby-KAN	Sentinel-2	94.5%	Burned area detection
2	Mahmoud et al.	2024	ML Change Detection	Optical	91.2%	Archaeological site monitoring
3	Singha et al.	2024	ML + RS + GIS	Multispectral	89.0%	Forest fire susceptibility
4	Prince	2024	CNN on Multispectral	Satellite Imagery	93.4%	Multi-spectral wildfire detection
5	Zhang et al.	2021	U-Net (SAR + Optical)	SAR + Optical	93.2%	Real-time wildfire monitoring
6	Sathishkumar et al.	2022	Deep Learning	Multispectral	90.0%	Smoke + fire detection
7	Moghim & Mehrabi	2024	RF, SVM, NB	RS Imagery	88.9%	Algorithm comparison across regions
8	Thị et al.	2024	Explainable Hybrid ML	Multisource	91.5%	Susceptibility mapping
9	Seddouki et al.	2023	SVM + RS	Landsat 8	87.0%	Forest fire mapping
10	Ahajjam et al.	2024	Ensemble ML + ST Clustering	Multi-source	93.7%	Wildfire Behavior prediction

11	Marjani et al.	2024	ASPP-enabled CNN	Sentinel-2	91.8%	Explainable wildfire modeling
12	Ban et al.	2020	Deep Learning	Sentinel-1	94.0%	Real-time wildfire progression
13	Galanis et al.	2021	CNN (DamageMap)	Post-wildfire	95.0%	Building-level damage mapping
14	Gupta et al.	2018	CNN + Satellite	Multihazard	89.6%	Operational damage detection
15	Gupta et al.	2020	Semi-supervised Learning	Satellite	91.0%	Minimal labeling approach
16	Elizaroshan & Raj Kumar	2024	DL Image Detection	Satellite	88.5%	Real-time classification
17	Wang et al.	2023	Deep Vision	RS Images	90.1%	Forest wildfire education tool
18	Priya & Vani	2019	CNN	RS Imagery	89.0%	Early forest fire detection
19	Phan & Nguyen	2018	DL + STS	Satellite	91.0%	Early wildfire detection
20	Toan et al.	2019	DL Hyperspectral	RS Imagery	89.4%	Wildfire classification
21	Seydi et al.	2022	Fire-Net (DL)	Satellite	92.0%	Active fire detection
22	Bouguettaya et al.	2022	UAV + CV + DL	UAV Imagery	93.3%	Aerial early detection

23	Govil et al.	2020	DL + Remote Cameras	Ground imagery	90.0%	Near real-time alerts
24	Shamsoshoara et al.	2021	CNN (FLAME Dataset)	UAV	88.8%	Controlled burn classification
25	Xu et al.	2020	Satellite + CNN	Multisource	90.4%	Building damage detection
26	GFDRR	2010	Manual + RS	Satellite + Field	-	Haiti Earthquake case study
27	Braik & Koliou	2023	DL + Mapping	Multihazard	91.0%	Post-disaster urban mapping
28	Kaur et al.	2022	Hierarchical Transformer	RS Imagery	92.5%	Damage severity classification
29	Zhang et al.	2021	BDANet (CNN)	RS Imagery	91.7%	Cross-attention for segmentation
30	Alisjahbana et al.	2024	DeepDamageNet	RS + multi-disaster	90.2%	Dual-step damage assessment
31	Shen et al.	2020	CDFF-Net	Side-looking Satellite	89.9%	Feature fusion model
32	Gupta & Shah	2020	RescueNet	RS Imagery	92.1%	Damage segmentation/classification
33	Chen et al.	2022	Multi-source RS	Nepal Earthquake	88.6%	Real case building damage

34	Kashtan & Hnatushenko	2023	ML Classification	Digital Imagery	86.5%	Ukraine case study
35	Hordiiuk & Hnatushenko	2017	Laplacian Filter + NN	VHR Imagery	84.0%	Urban structure damage
36	Zhao et al.	2020	Boundary-regularized CNN	Satellite	89.7%	Building footprint extraction
37	Shen et al.	2021	S2Looking Dataset	Satellite	-	Change detection dataset release
38	Dimassi et al.	2021	CNN	VHR Imagery	88.0%	Structural classification
39	Ashrapov et al.	2022	DL Monitoring	Satellite	87.3%	Construction activity tracking
40	Li et al.	2022	Literature Review	RS Optical	-	Overview of VHR building detection